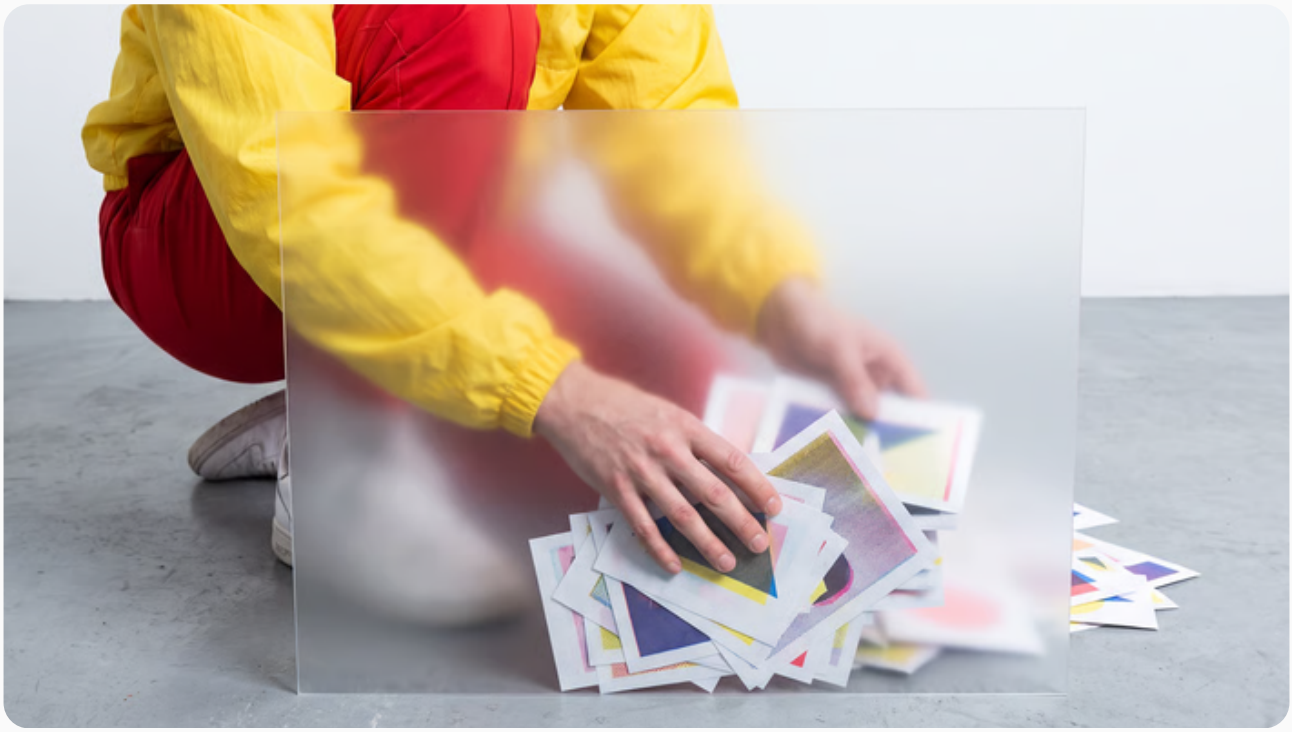
JEV RERANKER RAG RERANKING STEERABLE RERANKER

JEV as a Steerable Reranker: A Practical RAG Guide

How to wire JEV into a RAG pipeline as an instruction-following reranker, with comparisons against cross-encoders and LLM rerankers.

Edited by Luis Chavez-Mattos, Director of Product · September 21, 2026 · RSS



What is JEV and why does it matter for RAG?

JEV is a small, low-cost model built for making structured decisions rather than

generating free text. In a retrieval augmented generation pipeline, it slots into the points where a system has to decide something: is this chunk relevant, which of these candidates best matches a policy, how should these documents be ordered. It ships with three primitives: a binary true/false classifier (null), a choice ranker for picking among multiple options against a defined criteria, and a score ladder for assigning graded scores across a set of candidates. The reranking use case, which uses the choice ranker and scoring primitives together, is where JEV's design shows up most clearly, because reranking is really a repeated classification problem dressed up as a sorting problem.

TL;DR

- **JEV replaces frozen-in-time cross-encoders** with a reranker that can follow written instructions, so it adapts when business logic or policy changes without retraining.
- **The core primitives are binary classification, choice ranking, and score ladders**, and reranking in RAG mainly leans on the ranking and scoring primitives.
- **A criteria acts as an executable policy**: instead of a vague prompt, you give JEV strict decision boundaries (e.g., "true if the passage directly answers the question, false if it only shares keywords").
- **Changing the criteria changes the ranking**, which is the practical payoff: the same set of documents gets reordered differently depending on whether you ask JEV to favor an "authoritative" policy version or a "fast workaround" one.
- **JEV handles concurrency well**: sequential decisions that took around 40 seconds for a batch of queries dropped to roughly 7.6 seconds using parallel request pools.
- **Cost and latency scale with chunk size**, and JEV's advantage over LLM rerankers (tested against Gemini Flash and Flash-Lite in the demo) grows sharply as chunks get larger.
- **Adding JEV as a reranker on top of a BM25 baseline pushed top-1 retrieval accuracy from 21% to 54%** in the demonstrated benchmark, a meaningful jump for a single added step.



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How does JEV work as a reranker?

Traditional reranking with a cross-encoder takes a query and a candidate chunk, scores their relevance, and repeats across all candidates. Cross-encoders are fast and cheap, but they're static: they were trained once on some notion of relevance and can't be told to weigh things differently at inference time. If your organization changes a policy, but the underlying facts and phrasing in your documents stay similar, a cross-encoder has no way to know the older version should now rank lower.

JEV addresses this by taking a criteria alongside the query and candidates. That criteria functions as an explicit decision boundary: true if the passage matches condition X, false if it only shares surface-level keywords. Because the criteria is text you write and can change on demand, you can point JEV at "prefer the authoritative version 2 policy" for one run and "prefer the fast workaround approach" for another, and get two different valid rankings of the exact same document set. That steerability is the main practical difference from cross-encoders: the ranking behavior lives in the instructions you supply, not frozen in the model weights.

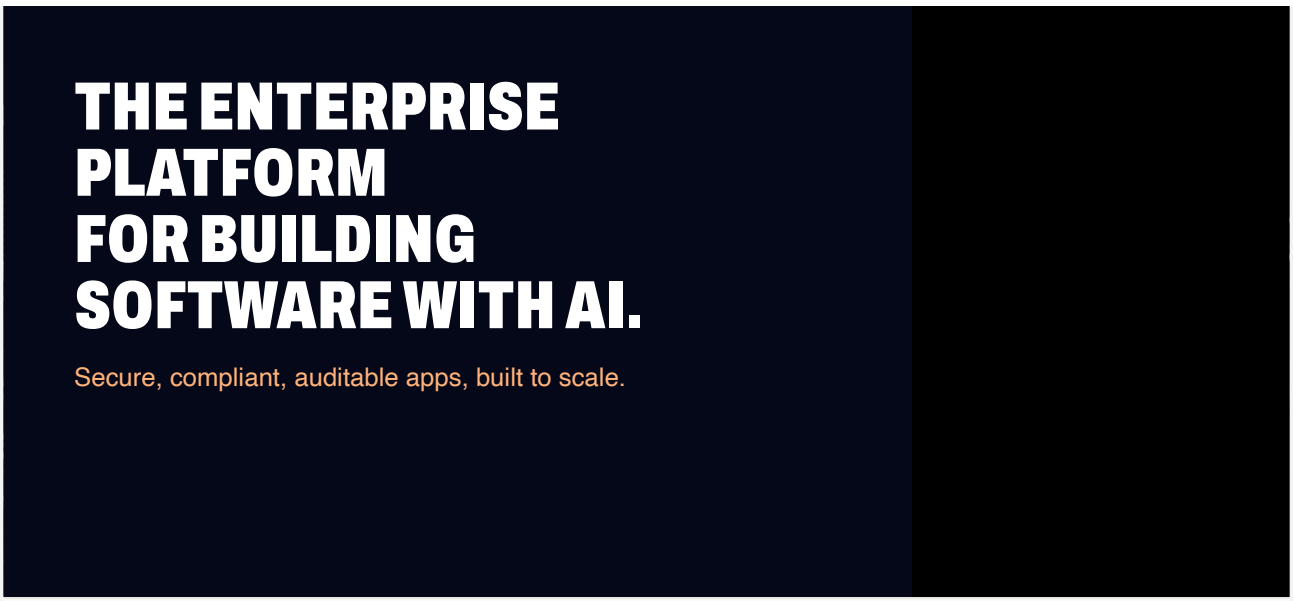
Compared to using a full LLM as a reranker (which can also follow instructions), JEV's advantage is cost and speed at scale. LLM reranking works, but pricing and latency climb fast, especially as chunk size grows, since you're paying for a full generative model to essentially output a relevance judgment. JEV is built specifically for that judgment task, so it stays closer in cost to a lightweight model like Gemini Flash-Lite at small chunk sizes and pulls further ahead as chunk sizes increase.

How do you set up a criteria-based reranking pipeline?

The practical workflow demonstrated follows a standard RAG shape with one addition:

1. **Retrieve candidates.** Pull chunks from vector search, full-text search (BM25 in the demo), or both.
2. **Define a criteria as policy.** Instead of a general prompt, write a strict rule: true if the passage directly answers the question being asked, false if it merely shares words or topics without answering it.
3. **Score each candidate against the criteria.** JEV assigns calibrated probabilities to each option under the null (true/false) primitive. In a test with four sample passages (a direct answer, a topic-overlapping non-answer, a keyword-only match, and an unrelated passage), JEV assigned the highest probability to the true direct answer and noticeably lower confidence to the rest.
4. **Set a threshold.** Because the output is a calibrated probability rather than a raw similarity score, you can pick a cutoff below which a chunk gets discarded and above which it's accepted, effectively turning JEV into a relevance filter as well as a ranker.
5. **Swap the criteria when policy changes.** Because the decision boundary is just text, updating it doesn't require retraining or re-embedding anything, only rewriting the instruction.





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This is the part that matters for anyone maintaining a RAG system over time: policies, documentation versions, and business rules drift. Vector similarity measures semantic closeness, not correctness under current rules. A reranker that can be re-pointed at a new policy without retraining closes that gap.

Is JEV worth it compared to cross-encoders and LLM rerankers?

For static ranking tasks where relevance never changes definition, a cross-encoder is still the cheaper, faster default; it doesn't need a criteria and it's a well-understood, mature technique. The problem is any RAG system that lives inside a real organization: policies update, terminology shifts, and "correct" answers today may not be correct next quarter even though the source documents look similar. Cross-encoders can't adapt to that without retraining.

LLM-as-reranker solves the adaptability problem but at real cost. In the tested comparison against Gemini Flash and Flash-Lite, JEV tracked closely with the cheaper Flash-Lite model on cost and speed at small chunk sizes, but the gap widened as chunk size increased, with LLM reranking costs increasing roughly linearly while JEV stayed comparatively flat. For pipelines processing large chunks or high query volumes, that difference compounds quickly.

The retrieval accuracy numbers make the case concretely: a BM25-only baseline in the demonstrated benchmark hit 21% top-1 accuracy. Adding JEV as a reranking step on top of that same retrieval pushed accuracy to 54%, without changing the retriever itself. That's a reranking layer doing meaningful work, not a marginal tweak, and it was measured at top-1, meaning top-K accuracy (which most production systems actually rely on) would likely be higher still.

Concurrency is the other practical factor. The demo reported around 17 decisions per second sequentially, which meant roughly 40 seconds for a batch of queries run one at a time. Using a pool of 64 concurrent requests brought that same batch down to about 7.6 seconds. For anyone reranking dozens of chunks per query across many simultaneous users, that throughput difference is what makes a criteria-based reranker usable in production rather than just in a notebook.

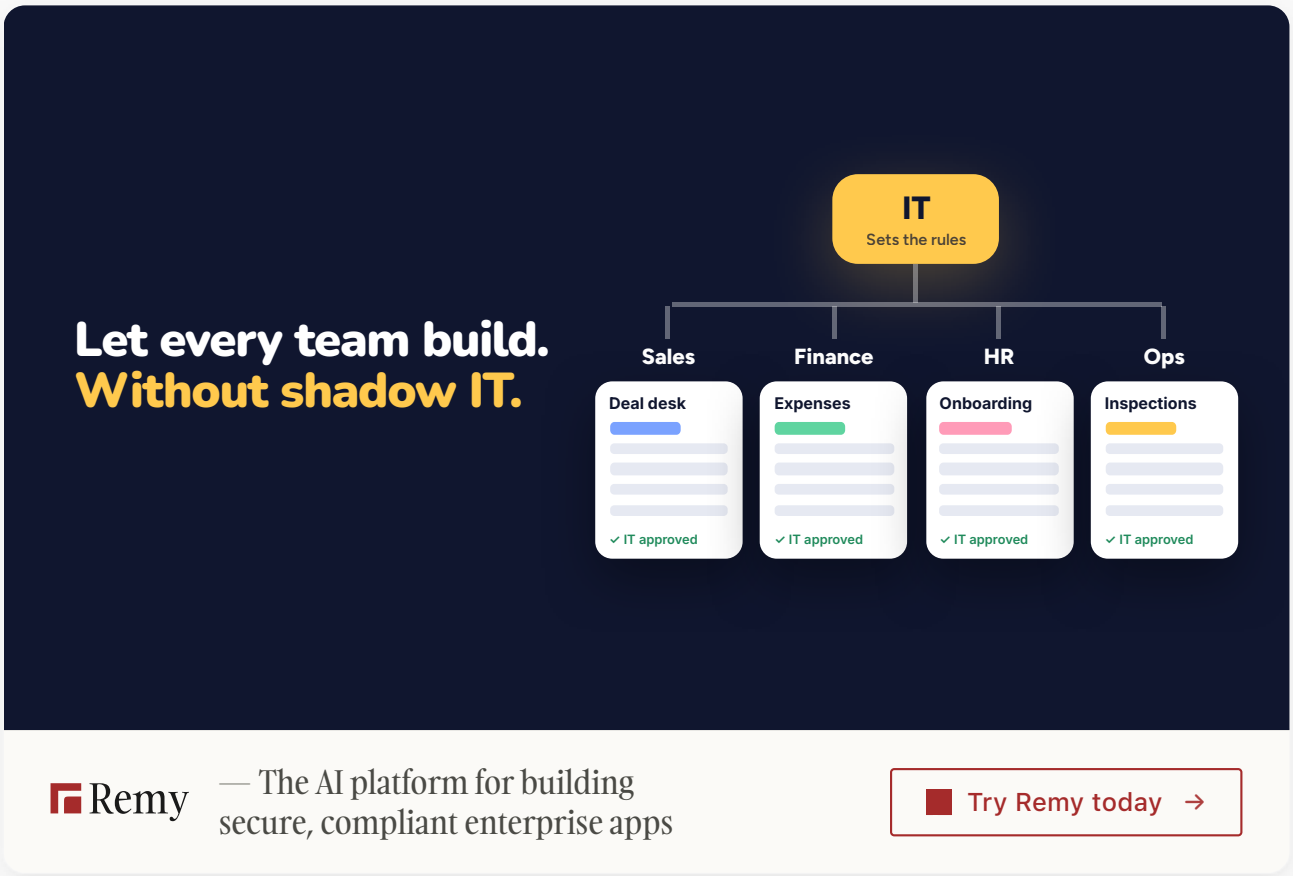
Where else can this kind of steerable classifier fit in a RAG pipeline?

Reranking is the clearest use case, but it's not the only decision point in a RAG system that benefits from an instruction-following classifier. Filtering retrieved chunks before they reach the generator is effectively the same binary classification task with a different threshold. Citation validation, checking whether a generated claim is actually supported by a specific retrieved passage, maps onto the same true/false criteria structure. Entity validation in graph-based RAG systems, where extracted entities need to be checked against a corpus for correctness, is another natural fit for the same primitives, since it's ultimately still a yes/no or multiple-choice decision under a defined rule.

The general pattern is: any point in a pipeline where you're currently using a fixed rule, a brittle regex, or an expensive LLM call just to get a yes/no or a ranked choice, is a candidate for a small steerable classifier instead.

Frequently Asked Questions

What makes JEV different from a standard cross-encoder reranker?



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A cross-encoder scores query-document pairs based on patterns learned during training and can't be redirected at inference time. JEV takes an explicit criteria alongside the query and documents, so you can change what "relevant" means (favoring a specific policy version, a specific answer format, etc.) without retraining anything.

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Is JEV cheaper than using an LLM as a reranker?

In the tested comparison against Gemini Flash and Flash-Lite, JEV's cost and speed stayed close to the cheaper Flash-Lite model for small chunks, and the gap grew substantially in JEV's favor as chunk size increased, since LLM reranking costs rose roughly linearly with chunk size.

How much does reranking actually improve retrieval accuracy?

In one benchmark using a BM25 full-text search baseline, top-1 accuracy was 21%. Adding JEV as a reranking step raised it to 54%, without any change to the underlying retrieval method.

Can JEV handle many queries at once?

Yes. Sequential processing handled about 17 decisions per second, taking roughly 40 seconds for a batch of queries. Using a pool of 64 concurrent requests cut that same workload down to about 7.6 seconds.

Beyond reranking, where else can JEV fit in a RAG system?

Any point in the pipeline that currently requires a yes/no or multiple-choice decision is a candidate: filtering irrelevant chunks before generation, validating whether a citation actually supports a claim, or verifying extracted entities in a graph RAG setup.

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